Thais Isabelle Parron Ruiz

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EDUCATION

University of South Florida - Tampa, FL — GPA: 3.87
Bachelor of Science in Mechanical Engineering — Expected Graduation Date: Dec 2026
Minor: Aerospace Engineering

EXPERIENCE

Engineering Tutor — **AUG 2024 – Present**
University of South Florida — *Tampa, FL*

- Tutor at the Engineering Student Success Center in Statics, Dynamics, Thermodynamics, Electrical Systems I, programming, Computational Methods, and Kinematics & Dynamics of Machinery.
- Assist an average of **2** students per hour across all supported subjects.

Vibration-Based Corrosion Detection Research — **MAY 2026 – Present**
University of South Florida — *Tampa, FL*

- Perform impact hammer testing on uncorroded and corroded **12**-inch plates using two end-mounted accelerometers to compare changes in modal response.
- Evaluate mode shapes and nodal patterns from **51** impact locations spaced **0.25** inches apart, with centerline excitation used to reduce torsional modes and support corrosion location analysis.

PFAS-Free Fog Harvesting Research — **AUG 2024 – DEC 2025**
University of South Florida — *Tampa, FL*

- Designed and tested a PFAS-free fog harvesting system using laser-textured aluminum surfaces with superhydrophobic and patterned superhydrophilic/superhydrophobic regions to improve droplet capture, transport, and collection.
- Integrated Peltier cooling and evaluated untreated, superhydrophobic, and patterned samples under controlled fog conditions, with the patterned surface achieving the highest collection at **6.884** mL.

ENGINEERING PROJECTS

Dual-Action Can Crusher Design — **JAN 2025 – MAY 2025**
Kinematics & Dynamics of Machinery Final Project

- Designed a dual-action can crusher in SolidWorks using a **6**-bar linkage mechanism to pre-dent cans horizontally before completing vertical crushing.
- Reduced required input force by approximately **40%**, from **36** to **21** pound**s**, through prototype testing with mechanical and electrical force gauges.

Ladder Logic Control System Project — **JAN 2026 – MAY 2026**
Mechanical Controls | LogixPro PLC Simulator

- Developed PLC ladder logic programs in LogixPro for door control, two-way traffic control, and conveyor/silo automation using relays, timers, counters, sensors, and output coils.
- Tested automated logic for motor control, traffic light timing, proximity-based box counting, reset conditions, interlocks, and status indicator lights.

Computational Methods with MATLAB — **JAN 2025 – MAY 2025**
Computational Methods

- Applied MATLAB to solve engineering problems in fluid mechanics and heat transfer using numerical methods, matrix systems, finite differences, regression, interpolation, and spline fitting.
- Compared theoretical and numerical results using plots, relative error analysis, and engineering assumptions such as laminar flow, round-off error, and heat conduction behavior.

Projectile Simulation Game — **MAR 2024 – APR 2024**
Programming Concepts

- Developed a MATLAB projectile motion simulator using user-defined launch height, target location, launch angle, velocity, and gravity conditions for Earth, Mars, and Jupiter.
- Created dynamic trajectory visuals, input validation, multiple attempts, and feedback for maximum height and distance from the target within a **2**-meter accuracy requirement.

Acoustic Guitar CAD Model — **AUG 2023 – DEC 2023**
Computer Aided Design and Engineering

- Modeled a full acoustic guitar assembly in SolidWorks from scratch, including detailed part drawings, material selection, exploded views, and manufacturing-style documentation.
- Designed a functional tuning mechanism using shaft-mounted worms and worm gears to adjust string tension through rotational motion.

SKILLS

Software: SOLIDWORKS, MATLAB, Simulink, LabVIEW, LogixPro PLC Simulator, Autodesk Fusion 360, Microsoft Office
Programming/Tools: Arduino, MATLAB scripting, PLC ladder logic, CAD modeling, 3D printing
Languages: English (fluent), Portuguese (native), Spanish (basic).
Certifications: SOLIDWORKS Associate, Mechanical Design and Additive Manufacturing